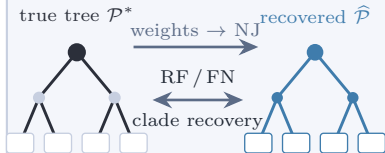


Fine-tuning leaves path-dependent structure in model weights. We estimate the resulting model phylogeny, score it against known training trees, and test whether weight proximity predicts behavioral similarity. We evaluate four-point additivity as a label-free diagnostic of clade recovery.

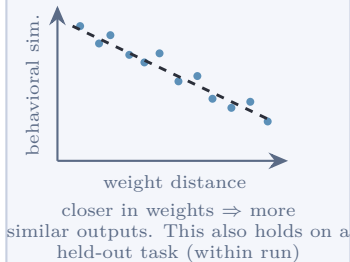
A. Controlled recovery (known truth)



neighbor-joining estimate,
scored against known truth

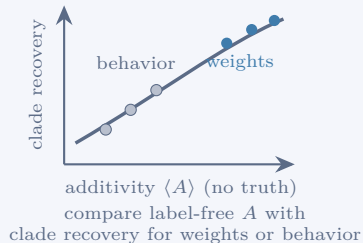
full weight access,
known lineage

B. Weights predict behavior random-effects pooled $r_{\text{DL}} < 0$



weight $\rightarrow$ behavior link

C. Additivity and recovery



diagnostic without known truth